Faultless Key Recovery

Iteration-Skip and Loop-Abort Fault Attacks on LESS

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Abstract. To enhance the diversity of basic hard problems underlying post-quantum cryptography (PQC) schemes, NIST launched an additional call for PQC signatures in 2023. Among numerous candidate schemes, several code-based ones, which have successfully advanced to the second round, are constructed by applying the Fiat–Shamir transform to the parallel repetition of a (relatively low soundness) commit-and-prove sigma protocol similar to the Stern identification scheme.

In Fiat–Shamir-based signatures, it is well-known that key material will be leaked if an attacker can somehow obtain what amounts, in the sigma protocol, to the responses to different challenges with respect to the same commitment. This idea is for example at the basis of a famous differential fault attack against deterministic Fiat–Shamir-based signatures like EdDSA. It is usually difficult to mount a fault injection attack based on that principle against a properly randomized Fiat–Shamir-based scheme however (at least with single faults): since commitment collisions are ruled out, it typically involves obtaining the responses to multiple challenges with respect to the same commitment within a single execution of the signature, which is often impossible by construction (e.g., because the extra information will not fit in a single signature, or because it is hard to force the computation of both responses).

Due to the comparative inefficiency of signatures based on Stern-like protocols with parallel repetition, candidate constructions are led to use clever compression techniques to reduce signature size, in a way that increases the attack surface for physical attacks. In this paper, we demonstrate this against the LESS signature scheme, which uses so-called GGM trees for signature compression. We propose a simple fault attack on the construction of a binary array used to build the GGM tree, and show that a small number of faulty signatures suffice for full key recovery.

We provide a thorough mathematical model of the attack as well as extensive experimental validation with glitch attacks on a ChipWhisperer board, showing that, depending on the target parameter set and the precise fault model we consider, full key recovery can very often be achieved with just one or two faulty signatures, and never more than a couple hundred even in the least favorable scenario for the attacker.

Keywords: Fault Analysis · Post-Quantum Cryptography · LESS Signature Scheme · ChipWhisperer

1 Introduction

In 2016, the National Institute of Standards and Technology (NIST) launched the post-quantum cryptography standardization project [CML17] and called for post-quantum cryptography (PQC) proposals. After 4 rounds of selection, Crystals-Kyber [SAB+22], Dilithium [LDK+22], FALCON [PFH+22], SPHINCS+ [HBD+22] and the later added scheme HQC [AAB+22] were selected for standardization.

Remarkably, three of those five selected schemes are lattice-based, leading to concerns regarding the repercussions of a hypothetical breakthrough in the cryptanalysis of lattice problems, structured or otherwise. Moreover, the selection process saw the elimination of certain candidates perceived as strong contenders, like Rainbow [DCP+20] and SIKE [JAC+20], at quite a late stage due to dramatic developments in the cryptanalysis of the respective underlying assumptions [Beu22, CD23, Rob23]. This made it desirable to try and further diversify the type of hard problems future post-quantum standards could be built upon. To do so, NIST launched an additional call for post-quantum signatures in 2023, that received numerous submissions based on a variety of related to multivariate equations, elliptic curve isogenies, codes, hash functions and more.

Code-based schemes in particular were well-represented among the candidates to this additional call, with the submission of three candidates: LESS [BBB+24b], CROSS [BBB+24a], and Wave [BCC+23] (LESS and CROSS have since progressed to the second round). This is interesting as, while code-based cryptography has long been a major source of encryption schemes, dating back to the McEliece cryptosystem [McE78] back in the infancy of public-key cryptography (and all the way to recent schemes like HQC), code-based signatures are far less common. Prior examples mostly include the Courtois–Finiasz–Sendrier (CFS) scheme [CFS01], some versions of which have faced significant security issues [FGO+13, PT16], and comparatively inefficient schemes obtained via the Fiat–Shamir transform from code-based identification protocols like Stern's [Ste94]. Wave can be seen as a rigorous approach to CFS-like hash-and-sign constructions, while LESS and CROSS are modern, optimized takes on the Fiat–Shamir approach.

This paper focuses on LESS, which relies on the hardness of the Linear Equivalence Problem (LEP) [SS13]. First proposed in [BMPS20], LESS was argued to be secure against both traditional and quantum adversaries, with specific resistance to the Leon [Leo82] and SSA [Sen99] attack algorithms. However, this original version was compromised shortly after by an improved Leon algorithm [Beu20]. To defend against this attack and reduce the signature size, an improvement [BBPS21] was proposed for the challenge phase, which introduced multiple secret keys and fixed the challenge weight. Later, the Information Set Linear Equivalence (IS-LEP) approach was proposed in [PS23] and adapted to LESS. This reduced communication costs by half by verifying the systematic form (SF) of non-pivot columns rather than the full matrix. These improvements were integrated into LESS-v1 [BBB+23], which was submitted to the first round of the additional call. To further optimize the signature size and accelerate the signing process, some new improvements have also been applied, including the introduction of "canonical form" (CF) in [CPS23], the optimized the calculation of the "row reduced echelon form" (RREF) and CF. Meanwhile, some new features like "pivot column reuse", "variable-time RREF", and "blinding techniques" discussed in [BEP+25] to further improve the issue in the scheme. These recent improvements are incorporated into LESS-v2 [BBB+24b] submitted to NIST.

Another notable requirement in the additional call is that NIST emphasized the significance of side-channel attack resistance and implementation robustness for candidate schemes. This is because even if a scheme is provably secure, physical attacks remain a significant threat to its security. This domain has been extensively explored for lattice-based candidates, with significant quantitative research carried out regarding their physical vulnerabilities [IMS+22, FKK+22, ZLYW23, UMB+23, UMB+23, GGHS25, AWK+25, HPT25]. In contrast, equivalent research efforts for code-based schemes remain scarce. Al-

though there have been several studies targeting McEliece [STM+08, PGD+22, GJJ22, BCM+23], physical attacks on signatures are few and far between: there appears to be only two published works related to LESS [MAKK24, CMR25] and one other work on CROSS [SS25].

Yet, there are reasons to believe that schemes like LESS and CROSS may have structural vulnerabilities with respect to physical attacks and particularly fault analysis. Indeed, as discussed earlier, they are constructed by applying the Fiat–Shamir transform [FS87] to underlying commit-and-prove identification protocols (or rather, since those protocols have relatively low soundness, to many parallel repetitions of such protocols). Such protocols have the "special soundness" property that, if one can obtain the prover's response to two different challenges with respect to the same commitment, one can directly extract the prover's secret key. This property makes Fiat–Shamir-based signatures with *deterministic* commitment randomness very vulnerable to differential fault analysis, as demonstrated in [ABF+18, PSS+18, BP18] against schemes like EdDSA [BDL+11] and deterministic variants of ECDSA or Dilithium. Properly randomized Fiat–Shamir-based scheme cannot be attacked directly in the same way, however (at least not with single faults): since commitment collisions are ruled out, it requires obtaining the responses to multiple challenges with respect to the same commitment within a single execution of the signature, which is often impossible by construction (e.g., because the extra information will not fit in a single signature, or because it is hard to force the computation of both responses). This should also be the case for schemes with parallel repetition that apply Fiat–Shamir in a straightforward way; but doing so to the underlying identification protocols of schemes like LESS and CROSS would result in very inefficient signatures, which is why those schemes are much more clever in the way they pack the responses into a single signature. LESS, for example, relies on a so-called GGM trees, making it less straightforward to ensure that overlapping responses will not be inadvertently released if the signature process is tampered with. Exploring the concrete security implications of that type of optimization is the main motivation of the current study, and may have broader ramifications beyond the specific case of the LESS signature schemes.

1.1 Our Contributions

In this paper, we proposed a fault attack on the latest LESS-v2 scheme. Compared with the existing attacks [MAKK24, MKN+25], our contributions can be summarized as follows.

Novel Fault Injection Target on LESS-v2. We propose a novel fault injection strategy at the `for` loop related to binary array generation before the seed tree construction in LESS-v2. This approach is significantly different from existing fault attack strategies which typically corrupt the complex seed tree structure. Unlike previous approaches, our method enables the direct derivation of the fault location from applied attack parameters, without guesses. By exploiting the target `for` loop's linearity and isochronicity, we simplify fault analysis to a binary determination of fault occurrence at a single, precisely controlled location, avoiding heuristic fault position inference within complex data structures. Compared to tree-structured fault injection methodologies, our attack provides greater flexibility in targeting fault injection positions. Once a successful fault injection is performed, the adversary only needs to retain the same attack parameters and shift the fault trigger time by an integer multiple of the clock cycles for one loop iteration, and a new fault can be triggered at other sequential locations with high probability. This flexibility mitigates timing variations that may cause high failure rates when repeatedly targeting fixed locations in real-world settings. Furthermore, this dynamic injection capability equips adversaries with a more adaptable strategy, especially when extra information about the signing program's internal state is available.

Comprehensive Fault Modeling. To make the attack practical, we analyzed the official C implementation of LESS-v2 across diverse hardware and operating systems and found that, although the target array is uninitialized in the C code, two distinct internal states occur with high probability: the zero-filled case, where the system consistently populates the array with zeros automatically, and the statically-addressed case, where the array maintains its value from previous execution contexts, resembling the behavior of a static array. We also consider two distinct fault models that are frequently discussed in the literature: the "Single Iteration Skip" model, where only one iteration in the target `for` loop is skipped; and the "Loop Abort" model, in which the injected fault causes all subsequent iterations of the loop to be skipped at once. In accordance with expectations and previous works, our tests and experiments confirm that, in practice, most common array element states are as described above, and triggering the two faults is not difficult.

Efficient Key Recovery Algorithm. We introduce a novel, two-phase algorithm to efficiently recover the secret monomial matrices making up the signing key. In this key recovery algorithm, the first phase focuses on recovering the permutation of $\mathbf{Q}_{i^*}$ using $2\log_2 n$ bits of leakage about it. By collecting these bits, we can extract intermediate values from the signing process without the key and retrieve the corresponding permutation via the set intersection method: each relevant leakage bit halves the size of the candidate set for specific position subsets in the permutation. Once the permutation is known, the second phase proceeds to completely recover the coefficients of $\mathbf{Q}_{i^*}$. This procedure successfully reconstructs an equivalent monomial matrix, $\mathbf{Q}'_{i^*}$, which is guaranteed to satisfy the crucial relationship $\mathrm{RREF}(\mathbf{G}_0(\mathbf{Q}'_{i^*})^{-1}) = \mathrm{RREF}(\mathbf{G}_0\mathbf{Q}_{i^*}^{-1})$. We prove that the recovered matrix $\mathbf{Q}'_{i^*}$ only differs from the actual secret monomial matrix $\mathbf{Q}_{i^*}$ by a non-zero scalar factor, making it effectively equivalent for all practical purposes. Even in the worst setting for the attacker, our results show that fewer than 300 faulty signatures suffice to recover all secret monomial matrices with high probability. In the Loop-Abort model, it is even the case that one or two faulty signatures are often sufficient.

Graph-Based Filtering via Disjoint Set Union. In the real world, fault injection attacks are unpredictable, indicating the success rate is not 100%. Consequently, when collecting signature results, valid faulty signatures generated by successful attacks are mixed with invalid signatures resulting from unsuccessful attempts. Thus, extracting valid information for key recovery from large-scale collected signature datasets is equally critical. To address this challenge, we propose a novel graph-based filtering algorithm leveraging the disjoint set union (DSU) data structure for the efficient validation of collected information. Our method is founded on verifying the consistency of information pairs that successfully yield partial secret permutations. This validation relies on two key properties: First, the candidate sets derived from valid information pairs must be mutually disjoint. Second, there must exist an exact, verifiable relationship between the size of these candidate sets and their observed frequency in the combined dataset. By leveraging the DSU data structure, our algorithm efficiently constructs connected components among all valid faulty signatures, enabling the rapid identification of the largest consistent data subset and the subsequent extraction of a vast majority of valid attack results.

1.2 Related Work

As mentioned previously, only a handful of papers so far appear to have explored physical attacks against the LESS and CROSS code-based schemes.

In passive side-channel analysis, [CMR25] proposed a comprehensive key-recovery attack on LESS and CROSS, exploiting the link between short-term and long-term secrets in secret vector-public matrix multiplication to extract the full private key from a single

power trace. Similarly, [SS25] designed a multi-stage CPA attack through a horizontal side-channel attack on CROSS. It targets at the syndrome calculation and achieves stepwise secret recovery with at most two power traces needed.

Regarding fault attacks, we are aware of only one published work on LESS-v1 [MAKK24] as well as a very recent preprint [MKN+25] applicable to LESS-v2.

The authors of [MAKK24] conducted a comprehensive fault analysis of LESS-v1 and CROSS. By conducting an attack against the seed-tree generation process, some internal nodes are leaked, which causes secret key leakage. However, that approach method requires guessing the fault location, and it does not apply to LESS-v2 due to significant structural differences between the two schemes. In contrast, our work targets the current scheme LESS-v2 and requires no fault location guessing.

The attack of [MKN+25] again targets the GGM-Tree construction process itself, and still involves fault location guessing. Moreover, it falls short of actual key recovery and only provides of way of forging signatures via fault injection. Unlike that work, our attack successfully recovers the secret key, and again does not involves fault location guessing.

2 Preliminaries

2.1 Notations

In this paper, we use $\mathbb{Z}_q = \{0, 1, ..., q-1\}$ to denote the minimum non-negative residue modulo q. Accordingly, we use $\mathbb{Z}_q^n$ and $\mathbb{Z}_q^{n\times m}$ to denote the set of n-dimensional vectors and $n \times m$ matrices over $\mathbb{Z}_q$, respectively. Throughout, vectors and arrays are denoted by lowercase bold letters (e.g., $\mathbf{v}$), and matrices by uppercase bold letters (e.g., $\mathbf{A}$). Also, we reserve specific symbols for dedicated use: calligraphic $\mathcal{C}$ denotes linear codes, and combinations is represented by upright C_n^m (e.g., $\mathrm{C}_4^2 = 6$).

Before introducing LESS and our attack methodologies, we also need to provide some extra definitions:

Definition 1 (Monomial matrix)**.** An $n \times n$ **monomial matrix** is a square matrix with exactly one nonzero entry in each row and each column. It has the form $\mathbf{Q} = \mathbf{PD}$ or $\mathbf{Q} = \mathbf{DP}$ where $\mathbf{P}$ is a permutation matrix and $\mathbf{D}$ is an invertible diagonal matrix. In this paper, we use M_n to represent the set of $n \times n$ monomial matrices over $\mathbb{Z}_q$.

Definition 2 (Linear code)**.** An (n, k)-linear code $\mathcal{C}$ over $\mathbb{Z}_q$ is a k-dimensional subspace of the vector space $\mathbb{Z}_q^n$. Equivalently, $\mathcal{C}$ can be described as the row space of a full-rank generator matrix $\mathbf{G} \in \mathbb{Z}_q^{k\times n}$. Each row vector of matrix $\mathbf{G}$ can be viewed as a codeword, and for any codeword $\mathbf{c} \in \mathcal{C}$, there exists a message vector $\mathbf{u} \in \mathbb{Z}_q^k$ such that $\mathbf{c} = \mathbf{uG}$.

Definition 3 (Equivalent linear code [BMPS20])**.** Two (n, k)-linear codes $\mathcal{C}_1, \mathcal{C}_2$ with corresponding generators $\mathbf{G}_1, \mathbf{G}_2 \in \mathbb{Z}_q^{k\times n}$ are said to be **equivalent** if there exists a monomial matrix $\mathbf{Q} \in \mathsf{M}_n$ and an invertible matrix $\mathbf{C} \in \mathbb{Z}_q^{k\times k}$ such that $\mathbf{G}_2 = \mathbf{CG}_1\mathbf{Q}$.

Definition 4 (Linear equivalence problem [SS13])**.** Fow two equivalent (n, k)-linear codes $\mathcal{C}_1, \mathcal{C}_2$ with their generators $\mathbf{G}_1, \mathbf{G}_2 \in \mathbb{Z}_q^{k\times n}$, find a solution matrix $\mathbf{Q} \in \mathsf{M}_n$ to satisfy the expression $\mathbf{G}_2 = \mathbf{CG}_1\mathbf{Q}$ in Definition 3.

Definition 5 (Disjoint set union [Tar75])**.** A **disjoint set union** (DSU) is a data structure to maintain a collection of pairwise disjoint dynamic sets $S_0, S_1, ..., S_{m-1}$, with $\bigcup_{i=0}^{m-1} S_i = U = \{0, 1, ..., n-1\}$. For any $x, y \in U$, it has the following functions:

- Query the set to which x belongs.
- Merge the sets containing x and y.

To support these functions, each set has a representative element, and it is maintained by the array $\mathsf{fa} : U \to U$ where $\mathsf{fa}_x = y$ denotes that the representation element of the set containing x is y. The key operations on array fa is as follows:

- $\mathsf{MakeDSU}(n)$: Initialize $\mathsf{fa}_i \leftarrow i$ for all $i = 0, 1, ..., n-1$, where each element forms an independent set initially.
- $\mathsf{Find}(x)$: Return the representative element of the set containing x:
  - If $\mathsf{fa}_x = x$, x is the representative element, and return x immediately.
  - If $\mathsf{fa}_x \neq x$, recursively query the value $\mathsf{Find}(\mathsf{fa}_x)$ until obtain a result. Then set $\mathsf{fa}_x \leftarrow \mathsf{Find}(\mathsf{fa}_x)$ to compress the query path and return fa_x as the result.
- $\mathsf{Union}(x, y)$: Merge the two sets containing x and y by setting $\mathsf{fa}_{\mathsf{Find}(x)} \leftarrow \mathsf{Find}(y)$.

2.2 Linear Equivalence Signature Scheme

The Linear Equivalence Signature Scheme (LESS) is adapted from the Sigma protocol (See Figure 1,2 in [BBB+23]) by iterating it t times through Fiat-Shamir transform [FS87]. In Sigma protocol, the signer can convince the verifier his/her knowledge to the secret matrix $\mathbf{Q} \in \mathsf{M}_n$ between two equivalent (n, k)-linear codes $\mathcal{C}_0, \mathcal{C}_1$ through repeated challenges in the zero-knowledge proof.

In LESS, the public parameters include the modulus $q = 127$, code size (n, k) with $n = 2k$, number of challenges t, number of non-zero challenges w, number of key matrices s, and public matrix $\mathbf{G}_0 = [\mathbf{I}_k \mid \mathbf{M}_{k\times k}] \in \mathbb{Z}_q^{k\times n}$. For theunified handling of zero and non-zero challenges, a dummy matrix $\mathbf{Q}_0 = \mathbf{I}_k$ is defined (not to be storaged). The signer's private key contains $s-1$ monomial matrices $\mathbf{Q}_1, \mathbf{Q}_2, ..., \mathbf{Q}_{s-1}$ with the public keys $\mathbf{G}_i = \mathrm{RREF}(\mathbf{G}_0\mathbf{Q}_i^{-1})$ for $i = 1, 2, \ldots, s-1$. For a challenge sequence $\mathbf{b} = (b_0, b_1, \ldots, b_{t-1})$, the signer constructs the response for each i: if $b_i \neq 0$, the response combines an intermediate nonce matrix $\tilde{\mathbf{Q}}_i \in \mathsf{M}_n$ and the private matrix $\mathbf{Q}_{b_i}$. Otherwise, only $\tilde{\mathbf{Q}}_i$ is needed.

To reduce the key and signature sizes, LESS uses seeds to store randomly generated values: the secret key is a seed $\mathsf{seed}_{\mathsf{priv}}$ with at most 64 bytes, from which all secret monomial matrices can be derived, while the public parameter $\mathbf{G}_0$ and intermediate nonce values $\tilde{\mathbf{Q}}_i$ are also seed-managed. For further signature size optimization, nonces $\tilde{\mathbf{Q}}_i$ $(i = 0, 1, \ldots, t-1)$ are organized into a GGM-Tree [GGM84], where each node stores a unique seed. The root seed is derived from $\mathsf{seed}_{\mathsf{priv}}$, and each child node's seed is deterministically generated by applying a cryptographic hash function to its corresponding parent node seed. By the hash function's one-wayness, the parent node seed remains computationally infeasible to recover even if all its child node seeds are published. For example, in Figure 1, with $t = 8, w = 4$ and $\mathbf{b} = (1, 0, 0, 0, 3, 1, 0, 2)$, the seeds of four nonce matrices $(\tilde{\mathbf{Q}}_1, \tilde{\mathbf{Q}}_2, \tilde{\mathbf{Q}}_3, \tilde{\mathbf{Q}}_6)$ should be published. However, with the GGM-Tree's management, only 3 seeds need to be published, as $\tilde{\mathbf{Q}}_2$ and $\tilde{\mathbf{Q}}_3$ can be derived from a common ancestor seed, and the verifier can reconstruct the tree to retrieve all required seeds.

Building on the above seed-based optimizations, Algorithm 1 outlines the signing process of LESS. Due to page limits, some preliminary steps are omitted. Here, the function `RREFWithTrack` returns both the RREF of the input matrix and a permutation matrix $\mathbf{T}_i$ (the inverse permutation of $\tilde{\mathbf{Q}}_i$). The function `CosetRep` extracts the first k entries of a permutation: for example, if $n = 8$ and $k = 4$, $\texttt{CosetRep}([1, 6, 4, 7, 0, 5, 3, 2]) = \{1, 4, 6, 7\}$. Thus, if $b_i = s^* \neq 0$ is the u-th non-zero challenge, the signer needs to return a k-element set $\mathsf{rsp}_u = \texttt{CosetRep}(\mathbf{Q}_{s^*}\mathbf{T}_i)$ instead of $\mathsf{seed}^{(i)}$. Otherwise, publishing $\mathsf{seed}^{(i)}$ would result in the leakage of sensitive information of $\mathbf{Q}_{s^*}$, which forms the basis of our attack.

Algorithm 1 LESS signing algorithm

Input: Public parameter (n, k, t, w, s), the public key seed length λ, private key seed $\mathsf{seed}_{\mathsf{priv}}$ (length: 2λ) and message msg

Output: The signature structure $\mathsf{sig} = (\mathsf{cmt}, \mathsf{salt}, \mathsf{path}, \mathsf{rep})$

Use $\mathsf{seed}_{\mathsf{priv}}$ to extract all the secret monomial matrices $\mathbf{Q}_1, \mathbf{Q}_2, ..., \mathbf{Q}_{s-1}$, the root seed of the GGM-Tree $\mathsf{seed}_{\mathsf{tree}}$.
Randomly sample the salt value with length 2λ.
Use $\mathsf{seed}_{\mathsf{tree}}$ to extract all the seeds for the nonce matrices: $\{\mathsf{seed}^{(0)}, \mathsf{seed}^{(1)}, ..., \mathsf{seed}^{(t-1)}\}$.
for $i = 0$ to $t - 1$ **do**
  $\tilde{\mathbf{Q}}_i \leftarrow \mathtt{SampleMonomial}(\mathsf{seed}^{(i)} \| \mathsf{salt} \| i)$
  $([\mathbf{I}_k | \mathbf{A}_i], \mathbf{T}_i) \leftarrow (\mathtt{RREFwithTrack}(\mathbf{G}_0 \tilde{\mathbf{Q}}_i))$
  $\mathbf{B}_i \leftarrow \mathrm{CF}(\mathbf{A}_i)$ //The "blind" operation on $\mathbf{A}_i$ is omitted here as it has no effect on $\mathbf{B}_i$
end for
$\mathsf{cmt} \leftarrow \mathrm{Hash}(\mathbf{B}_0 \| \mathbf{B}_1 \| ... \| \mathbf{B}_{t-1} \| \mathsf{salt} \| \mathsf{msg})$
$\mathbf{b} = (b_0, b_1, ..., b_{t-1}) \leftarrow \mathtt{SampleChallenge}(\mathsf{cmt})$
Create array $\mathbf{f} = (f_0, f_1, ..., f_{t-1})$ where $f_i \leftarrow 0$ if $b_i = 0$ else 1.
$\mathsf{path} \leftarrow \mathtt{GGMPath}(\mathbf{f})$ // store the seed to be published into array path
$j \leftarrow 0$
for $i = 0$ to $t - 1$ **do**
  if $b_i \neq 0$ **then**
    $\mathsf{rsp}_j \leftarrow \mathtt{CosetRep}(\mathbf{Q}_{b_i} \mathbf{T}_i)$ //rsp_j is a k-element set.
    $j \leftarrow j + 1$
  end if
end for
$\mathsf{rsp} \leftarrow (\mathsf{rsp}_0, \mathsf{rsp}_1, ..., \mathsf{rsp}_{w-1})$ //rsp is an array containing w k-element sets .
return $\mathsf{sig} \leftarrow (\mathsf{cmt}, \mathsf{salt}, \mathsf{path}, \mathsf{rsp})$.

3 Attack Point and Fault Models

3.1 Technical Overview

In the signing algorithm of LESS, we observed that between sampling array $\mathbf{b}$ and executing `GGMPath`, there exists an intermediate step in which array $\mathbf{b}$ is transformed into array $\mathbf{f}$. Furthermore, it is not array $\mathbf{b}$ but array $\mathbf{f}$ determines the seed publishment in LESS's implementation. In the official C implementation of LESS (Listing 1), array $\mathbf{f}$ is calculated from array $\mathbf{b}$ through a simple `for` loop.

Therefore, if the `for` loop in Listing 1 gets perturbed, some iterations may fail to execute, leading to specific elements in array $\mathbf{f}$ unmodified. As an uninitialized temporary

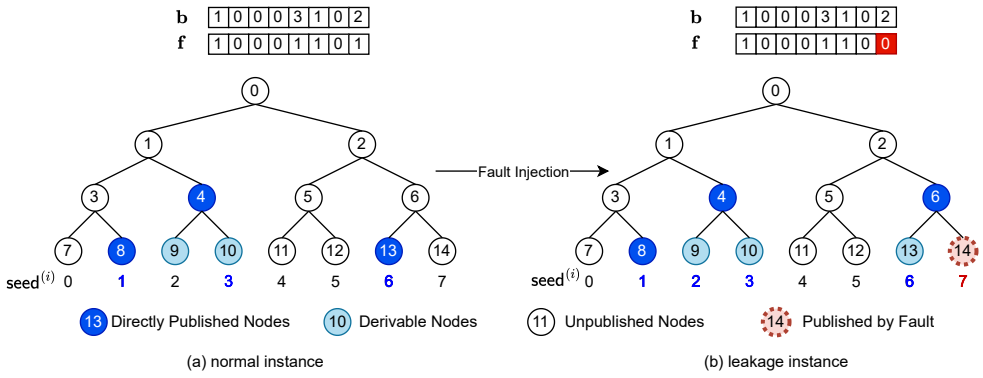


Figure 1: GGM tree structure and illustration of our fault injection

array, the content of array $\mathbf{f}$ should be random values strictly according to C standard. However, modern operating systems often incorporate memory reclamation and page zeroing strategies [YBF+11, AMH+16, LKS+17, SS18]. As a result, each element in array $\mathbf{f}$ has a high probability of being set to zero or even the whole $\mathbf{f}$ is zerofilled.

Meanwhile, we noticed all public parameters like n, t, w, s are defined as macros and determined during compilation, and all arrays defined in the signing function are fixed-length arrays. This deterministic execution and memory occupation mode results in a consistent stack frame layout of the signing function throughout the entire program runtime, and the memory allocation addresses of its internal temporary arrays exhibit high repeatability on some systems. Consequently, array $\mathbf{f}$ is prone to retaining residual values from its last invocation of the signing function, thereby creating a potential attack approach for adversaries.

Building upon the analysis, we compiled the official C implementation of LESS for all the scheme parameters across different compilers, devices, and operating systems. Our experiments confirm that the initial values of array $\mathbf{f}$ align with one of our expectations at most of the time, indicating the array's state can be derived under specific conditions. Therefore, we can perturb the `for` loop through fault injection attack to trigger some $b_{i^*} \neq 0$ with corresponding $f_{i^*} = 0$. With the knowledge of the faulty $\mathbf{f}$, we can reconstruct the GGM-Tree to to obtain the seed for the nonce matrix $\tilde{\mathbf{Q}}_{i^*}$, thereby gaining partial information about the secret monomial matrix $\mathbf{Q}_{b_{i^*}}$ to recover it (See Section 4).

For instance, as conceptually illustrated in the right subfigure in Figure 1: A fault is injected at index $i = 7$, causing $f_7 = 0$ with $b_7 = 2 \neq 0$. This triggered the leakage of the nonce matrix $\tilde{\mathbf{Q}}_7$. Through calculating $\mathbf{T}_7$ from $\tilde{\mathbf{Q}}_7$, some knowledge of the secret monomial matrix $\mathbf{Q}_{b_7} = \mathbf{Q}_2$ can be obtained, and the adversary can repeat the attack to obtain more knowledge until recovering full $\mathbf{Q}_2$ or even other secret $\mathbf{Q}$ values. Since a successful fault injection attack can result in multiple cases with $b_i \neq 0$ and corresponding $f_i = 0$, we provide the following two definitions:

Definition 6 (Leakage)**.** In the signing process, if a fault occurs where $b_i \neq 0$ but $f_i = 0$, we call it a **leakage**. A leakage means a seed should not be published gets published by fault, which leaks information about the corresponding secret monomial matrix. Specifically, if $b_i = s^* \neq 0$, we call it a leakage related to the secret monomial matrix $\mathbf{Q}_{s^*}$.

Definition 7 (Signing Oracle Access)**.** For an adversary performing the attack, we define a **signing oracle access** (or simply access) as the adversary obtains the output result of the signing oracle during the attack process. Therefore, a single signing oracle access may contain one or more leakages, or contain no leakage due to a failed attack or the adversary did not perform any attacks, but it will always include a (maybe) faulty signature provided by the signing oracle.

```
1  uint8_t fixed_weight_string[T]={0}; //Array b
2  SampleChallenge(fixed_weight_string, sig->digest); //b=SampleChallenge(cmt)
3
4  uint8_t indices_to_publish[T]; //Array f
5  for (uint32_t i = 0; i < T; i++) {
6      indices_to_publish[i] = !!(fixed_weight_string[i]);
7  }
8  // build GGM tree according to the array indices_to_publish
```

Listing 1: LESS.c implementation code snippet (lines 227-234)

3.2 Fault Models

Before intorducing our fault models, we first establish our assumptions for our fault analysis, which align with realistic adversarial capabilities in practical cryptanalytic scenarios:

- The private key in the signing oracle is deeply embedded and cannot be directly obtained by scanning its internal state. For each signing request, the signing oracle uses the fixed private key but employs a random salt value to sign the given message.
- The adversary has full access to the signing oracle and possesses complete public knowledge of the LESS scheme parameters and perform fault attacks on it. The adversary can also obtain a copy of the signing oracle environment without the secret key installed for debugging. However, the adversary does not know the private key or any secret internal values of the oracle.

Also, we provide a formal description here of two common cases of $\mathbf{f}$ mentioned in Section 3.1 to standardize the subsequent notation.

Case 1: Zero-Filled Array. In this case, array $\mathbf{f}$ is fully initialized with zero values automatically by the operating system. This behavior stems from the memeory zeroing in the operating-system level or compiler-enforced security mechanisms designed to prevent residual data leakage. In our test, when the signing function is first invoked, this case occurs across mainstream environments, including Raspberry Pi OS and Kali Linux 24Q4 with the g++ compiler, as well as Ubuntu 24.04, 25.04, and 25.10 with the clang compiler. This case mostly applies when the server spawns a child process for signature generation upon receiving a signing request and the child process exits immediately after producing the signature, with each request independently triggering a new child process invocation.

Case 2: Statically-Addressed Array. In this case, array $\mathbf{f}$ retains its residual values from prior invocations of the signature function, as its memory address in the stack frame of the signing procedure remains fixed across repeated calls. The initial state of $\mathbf{f}$ does not have to be most or all zeros, and only determined by the last values written to its fixed memory region. In our tests, this case can be detected on Windows 10 19H2 with MinGW g++ compiler somtimes. This case is suited for applications with a resident signature program, where the program runs continuously and invokes the signature function immediately upon receiving a signing request.

However, even in the two cases above, the state of $\mathbf{f}$ remains unobtainable due to the destructive nature of fault injection. Therefore, a refined attack must minimize the impact on the target's execution. For this purpose, we define the following fault models to ensure that the state of $\mathbf{f}$ is obtainable after a successful attack, enabling correct GGM-Tree reconstruction and seed extraction.

Fault Model 1: Single Iteration Skip. In this model, the adversary performs the attack to skip a specific iteration (e.g., the i^*-th iteration) in the `for` loop illustrated in Listing 1, and leave other iterations executed normally. This fault blocks the assignment $f_{i^*} \leftarrow !!(b_{i^*})$ from occurring, leaving f_{i^*} in its initial state. If $b_{i^*} \neq 0$ and the initial value of f_{i^*} is 0, then the adversary obtains a leakage related to $\mathbf{Q}_{b_{i^*}}$. Since the fault only affects one element in $\mathbf{f}$, at most one leakage can be obtained per single access.

Fault Model 2: Loop Abort. In this model, the `for` loop terminates **after** the i^*-th iteration with previous iterations executed normally. It is equivalent to inserting a `break` statement at the end of the i^*-th iteration or bypassing the loop's jump-back instruction

at the assembly level. This fault leaves all elements $f_{i^*+1}, f_{i^*+2}, ..., f_{t-1}$ unmodified. Compared with the "Single Interation Skip" model, more than one leakage might be occured through one access to the signing oracle.

4 Fault Injection Attack on LESS

4.1 Key Recovery

Our key recovery process consists of two steps: the permutation recovery and the coefficients recovery. For convenience, in this subsection, we assume all the attack trials are successful, and array $\mathbf{f}$ can also be obtained to rebuild the GGM-Tree and retrieve the seeds for the target nonce matrices.

4.1.1 Step 1: Permutation Recovery

To recover the permutation of the s^*-th private monomial matrix $\mathbf{Q}_{s^*}$, we first extract the challenge array $\mathbf{b}$ from the cmt value in the signature. Then we fill the fault-injected array $\mathbf{f}$ according to the target index and the fault model to rebuild the GGM-Tree and extract the leaf seeds. Through this procedure, we can save all the information of leakages related to $\mathbf{Q}_{s^*}$. By exploiting the leaked nonce $\tilde{\mathbf{Q}}_i$, the permutation matrix $\mathbf{T}$ and its corresponding permutation $\mathbf{p} = (p_0, p_1, ..., p_{n-1})$ inside the signing algorithm can be obtained. Then we form a set V containing the first k values in permutation $\mathbf{p}$ and collect its corresponding set R from the set array rsp in the faulty signature. Such a pair (V, R) leaks some information about the permutation of $\mathbf{Q}_{s^*}$ as $R = \mathbf{p}_V = \{p_v \mid v \in V\}$. The detailed extracting process is described in Algorithm 2.

Algorithm 2 Extract the leakage information related to $\mathbf{Q}_{s^*}$ from the faulty signatures

Input: Public parameter (n, k, t, w, s) with $n = 2k$, target key id s^*, faulty signature $\mathsf{fsig} = \{\mathsf{cmt}, \mathsf{salt}, \mathsf{path}, \mathsf{rsp}\}$, and faulty indexes $L = \{\ell_0, \ell_1, \ldots, \ell_{m-1}\}$
Output: Permutation set V and corresponding response $R = \mathsf{fsig.rsp_u}$ associated with $\mathbf{Q}_{s^*}$

$\mathbf{b} = (b_0, b_1, \ldots, b_{t-1}) \leftarrow \mathtt{SampleChallenge}(\mathsf{fsig.cmt})$
Fill the array $\mathbf{f}$ based on the faulty position and the predefined error model.
$\mathsf{GGMTree} \leftarrow \mathtt{RebuildGGMTree}(\mathsf{fsig.path}, \mathbf{f})$
$\{\mathsf{seed}^{(i)}_{\mathrm{leaf}}\} \leftarrow \mathtt{TreeLeaves}(\mathsf{GGMTree})$
for $u = 0$ to $t - 1$ **do**
  if $(b_u = s^*) \wedge (\mathbf{f}_u = 0)$ **then**
    $([\mathbf{I}_m \mid \mathbf{A}], \mathbf{T}) \leftarrow \mathtt{RREFWithTrack}(\mathbf{G}_0 \tilde{\mathbf{Q}}_{\ell_u})$
    Convert the permutation matrix $\mathbf{T}$ to the permutation sequence $\mathbf{p} = (p_0, p_1, \ldots, p_{n-1})$.
    $V \leftarrow \{p_0, p_1, \ldots, p_{k-1}\}$
    $z \leftarrow$ the number of non-zero values in $\{b_0, b_1, \ldots, b_{u-1}\}$
    Save the set V and $R = \mathsf{fsig.rsp}_z$ to the output file as the input of Algorithm 3.
  end if
end for

After collecting enough (V, R) pairs, we can employ our set intersection method to recover the permutation of the secret monomial matrix $\mathbf{Q}_{s^*}$, which is precisely defined in Algorithm 3.

However, Algorithm 3 is notably vulnerable to irrelevant data from failed attacks as any (V, R) pair not related to $\mathbf{Q}_{s^*}$ can trigger its returning $\perp$. Therefore, it is critical to filter related (V, R) pairs from all collected data for the recovery of $\mathbf{Q}_{s^*}$. We detail the filtering algorithm for removing all irrelevant (V, R) pairs comprehensively in Section 4.4.

Algorithm 3 Recover permutation of secret key $\mathbf{Q}_{s^*}$ with set intersection method

Input: Public param (n, k, t, w, s),the number of the collected data m,the permutations $\{V_0, V_1, ..., V_{m-1}\}$ and their corresponding response $\{R_0, R_1, R_2, ..., R_{m-1}\}$
Output: Permutation matrix $\mathbf{P}$ or $\perp$

Initialize $\mathsf{Candidate}_i \leftarrow \mathbb{Z}_n$ for $i = 0, \ldots, n-1$ // $\mathsf{Candidate}_i$: set, $\mathsf{Candidate}$: set array.
for $u = 0$ to $m-1$ **do**
  for $i = 0$ to $n-1$ **do**
    if $i \in R_u$ **then**
      $\mathsf{Candidate}_i \leftarrow \mathsf{Candidate}_i \cap V_u$
    else
      $\mathsf{Candidate}_i \leftarrow \mathsf{Candidate}_i \cap V_u^c$
    end if
    if $\mathsf{Candidate}_i = \emptyset$ **then**
      return $\perp$
    end if
  end for
  if All $|\mathsf{Candidate}_i| = 1$ and $(\bigcup_{i=0}^{n-1} \mathsf{Candidate}_i) = \mathbb{Z}_n$ **then**
    Record the unique element in each $\mathsf{Candidate}_i$ in order as permutation sequence $\mathbf{p}'$.
    return The permutation matrix $\mathbf{P}$ corresponding to the sequence $\mathbf{p}'$.
  end if
end for
return $\perp$

4.1.2 Step 2: Coefficients Recovery

With Algorithms 2 and 3, we can recover the permutation of a specific private monomial matrix $\mathbf{Q}_{s^*}$. By iterating s^* from 1 to $s-1$, the permutations of all the private monomial matrices can be recovered. However, to recover the real secret matrices, coefficient recovery is also demanded. Without loss of generality, we still take the coefficient recovery of the s^*-th private monomial matrix as an example.

First, according to Definition 1, the monomial matrix $\mathbf{Q}_{s^*}$ has the form $\mathbf{Q}_{s^*} = \mathbf{DP}$. Then we have

$$\mathbf{G}_{s^*}^{\mathtt{raw}} = \mathbf{G}_0 \mathbf{Q}_{s^*}^{-1} = \mathbf{G}_0 \mathbf{P}^{-1} \mathbf{D}^{-1} \tag{1}$$

Since $\mathbf{G}_0$ and $\mathbf{P}$ are known, while $\mathbf{Q}_{s^*}$ and $\mathbf{D}$ remain unknown, we can compute an auxiliary value $\mathbf{H}^{\mathtt{raw}}$ by treating $\mathbf{G}_0\mathbf{P}^{-1}$ as a single entity:

$$\mathbf{H}^{\mathtt{raw}} = \mathbf{G}_0 \mathbf{P}^{-1} \tag{2}$$

As $n = 2k$, the matrix $\mathbf{H}^{\mathtt{raw}}$ can be regarded as the column-wise concatenation of two square matrices $\mathbf{B}, \mathbf{C} \in \mathbb{Z}_q^{k \times k}$, i.e., $\mathbf{H}^{\mathtt{raw}}$ admits the form:

$$\mathbf{H}^{\mathtt{raw}} = [\mathbf{B}|\mathbf{C}] \tag{3}$$

Therefore, performing RREF on $\mathbf{H}^{\mathtt{raw}}$ is essentially equivalent to computing

$$\mathbf{H} = \mathrm{RREF}(\mathbf{H}^{\mathtt{raw}}) = \mathrm{RREF}([\mathbf{B}|\mathbf{C}]) = \mathbf{B}^{-1}[\mathbf{B}|\mathbf{C}] = [\mathbf{I}|\mathbf{B}^{-1}\mathbf{C}] \tag{4}$$

Substituting (3) into (1), we obtain:

$$\mathbf{G}_{s^*}^{\mathtt{raw}} = [\mathbf{B}|\mathbf{C}]\mathbf{D}^{-1} \tag{5}$$

Given that $\mathbf{D}$ is an $n \times n$ diagonal matrix, it can be partitioned into a block diagonal structure comprising $\mathbf{D}_1$ and $\mathbf{D}_2$, both of which are $k \times k$ diagonal matrices, i.e.:

$$\mathbf{D} = \begin{bmatrix} \mathbf{D}_1 & \mathbf{0} \\ \mathbf{0} & \mathbf{D}_2 \end{bmatrix} \tag{6}$$

Substituting (6) into (5), we obtain:

$$\mathbf{G}_{s^*}^{\mathtt{raw}} = [\mathbf{B}|\mathbf{C}] \begin{bmatrix} \mathbf{D}_1^{-1} & \mathbf{0} \\ \mathbf{0} & \mathbf{D}_2^{-1} \end{bmatrix} = [\mathbf{B}\mathbf{D}_1^{-1}|\mathbf{C}\mathbf{D}_2^{-1}] \tag{7}$$

Therefore, it can be derived that

$$\mathbf{G}_{s^*} = \mathrm{RREF}(\mathbf{G}_{s^*}^{\mathtt{raw}}) = (\mathbf{B}\mathbf{D}_1^{-1})^{-1}[\mathbf{B}\mathbf{D}_1^{-1}|\mathbf{C}\mathbf{D}_2^{-1}] = [\mathbf{I}|\mathbf{D}_1\mathbf{B}^{-1}\mathbf{C}\mathbf{D}_2^{-1}] \tag{8}$$

Let the right halves of $\mathbf{H}$ and $\mathbf{G}_{s^*}$ be $\mathbf{H}_{\mathrm{R}}$ and $\mathbf{G}_{\mathrm{R}}$ respectively, then we have:

$$\mathbf{G}_{\mathrm{R}} = \mathbf{D}_1\mathbf{H}_{\mathrm{R}}\mathbf{D}_2^{-1} \tag{9}$$

Therefore, for $\mathbf{H}_{\mathrm{R}} = [h_{ij}]$, $\mathbf{G}_{\mathrm{R}} = [g_{ij}]$, $\mathbf{D}_1 = \mathrm{diag}\left(d_{(1,0)}, d_{(1,1)}, ..., d_{(1,k-1)}\right)$ and $\mathbf{D}_2 = \mathrm{diag}\left(d_{(2,0)}, d_{(2,1)}, ..., d_{(2,k-1)}\right)$, we have:

$$g_{ij} = \frac{d_{(1,i)}h_{ij}}{d_{(2,j)}} \tag{10}$$

Herein, we define the matrix $\mathbf{U} = [u_{ij}] \in \mathbb{Z}_q^{k\times k}$ with $u_{ij} = \frac{g_{ij}}{h_{ij}}$. Specifically, if $h_{ij} = 0$, we define $u_{ij} = 0$. Theoretically, in the absence of such singular cases, $\mathbf{U}$ is expected to be a rank-1 matrix with no zero elements. Nevertheless, since both $\mathbf{G}_{\mathrm{R}}$ and $\mathbf{H}_{\mathrm{R}}$ are defined over the modular field $\mathbb{Z}_q$, some $g_{ij} = 0$ or $h_{ij} = 0$ disrupted the inherent rank-1 structure of $\mathbf{U}$. Notably, even in the presence of such zeros, the ratio between the non-zero elements in the corresponding rows or columns of $\mathbf{U}$ remains invariant. Therefore, we can still attempt to recover the proportional relationships between any two rows and between any two columns of $\mathbf{U}$, with all zero entries omitted in the calculation.

For each set of row vectors $(\mathbf{u}_0, \mathbf{u}_1, \ldots, \mathbf{u}_{k-1})$ and column vectors $(\mathbf{u}'_0, \mathbf{u}'_1, \ldots, \mathbf{u}'_{k-1})$ of $\mathbf{U}$, we formally define the zero-omitted ratios between any two rows or two columns using the notational forms $\frac{\mathbf{u}_i}{\mathbf{u}_j}$ and $\frac{\mathbf{u}'_i}{\mathbf{u}'_j}$ respectively. Then, we can compute:

$$\tilde{\mathbf{d}_1} = \left(\tilde{d}_{(1,0)}, \tilde{d}_{(1,1)}, ..., \tilde{d}_{(1,k-1)}\right) = \left(\frac{\mathbf{u}'_0}{\mathbf{u}'_0}, \frac{\mathbf{u}'_1}{\mathbf{u}'_0}, ..., \frac{\mathbf{u}'_{k-1}}{\mathbf{u}'_0}\right) \tag{11}$$

and

$$\tilde{\mathbf{d}_2} = \left(\tilde{d}_{(2,0)}, \tilde{d}_{(2,1)}, ..., \tilde{d}_{(2,k-1)}\right) = \left(\frac{\mathbf{u}_0}{\mathbf{u}_0}, \frac{\mathbf{u}_1}{\mathbf{u}_0}, ..., \frac{\mathbf{u}_{k-1}}{\mathbf{u}_0}\right) \tag{12}$$

After obtaining $\tilde{\mathbf{d}_1}$ and $\tilde{\mathbf{d}_2}$, we can derive the coefficient matrix $\mathbf{D}_{s^*} = \mathrm{diag}(d_0, d_1, \ldots, d_{n-1})$ $(0 \le i \le k-1)$ through the following relationship:

$$\tilde{d}_{(2,i)} = \frac{d_i}{d_0}, \quad \tilde{d}_{(1,i)} = \frac{d_k}{d_{k+i}} \tag{13}$$

Therefore, the solution space can be determined by fixing $d_0 = 1$ with enumerating all possible values of d_k, and generating the corresponding candidate matrices $\mathbf{D}'$ and the corresponding private key candidates $\mathbf{Q}'_{s^*}$. We then validate each candidate by computing $\mathrm{RREF}(\mathbf{G}_0(\mathbf{Q}'_{s^*})^{-1})$ and comparing the result to $\mathbf{G}_{s^*}$. If $\mathbf{G}_{s^*} = \mathrm{RREF}(\mathbf{G}_0(\mathbf{Q}'_{s^*})^{-1})$ holds, $\mathbf{Q}'_{s^*}$ is a valid solution that is equivalent to the secret matrix in the actual private key. This equivalence stems from the scale invariance of the RREF operation: for any non-zero scalar $a \in \mathbb{Z}_q$, $\mathrm{RREF}(\mathbf{G}) = \mathrm{RREF}(a\mathbf{G})$, and $\mathbf{Q}'_{s^*}$ differs from the real $\mathbf{Q}_{s^*}$ only by a constant scalar factor.

4.2 Clock Glitch Attack Implementation

The clock glitch attack is a fault injection technique that induces momentary perturbations in the target device's clock signal. These perturbations cause transient timing violations, which can result in instruction skipping or disturbance in program execution.

We implemented the clock glitch attack using the ChipWhisperer toolkit [OC14], with the ChipWhisperer Lite as the capture scope and the CW308 UFO board equipped with an STM32F303 microcontroller as the target device.

According to the ChipWhisperer documentation, the attack parameters used in the experiment are defined by the triple $(\mathsf{ext\_offset}, \mathsf{width}, \mathsf{offset})$:

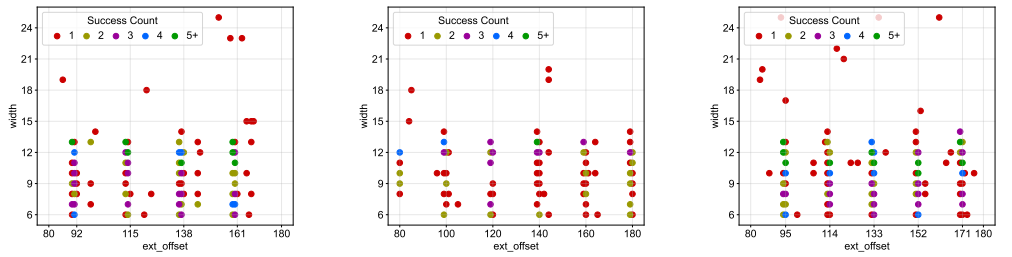
- $\mathsf{ext\_offset}$: A positive integer representing the number of clock cycles between the execution of the `trigger_high` flag and the initiation of glitch injection.
- width: A real number in $[0, 50]$, denoting the glitch pulse width as a percentage of the full clock cycle duration.
- offset: A real number in $[-48.8, +48.8]$, specifying the percentage adjustment of the glitch trigger position within a single clock cycle. The cumulative glitch trigger point is calculated as $\mathsf{ext\_offset} + 0.01 \times \mathsf{offset}$ clock cycles after `trigger_high`.

4.3 Finding the Proper Attack Parameter

To conduct an effective fault attack experiment via the ChipWhisperer platform, we integrated the `trigger_high()` and `trigger_low()` functions before and after the target loop. Furthermore, to verify whether the configured attack parameters achieve the expected results, we inserted an extra loop to recalculate array $\mathbf{f}$ and validate the effect. This supplementary check has no impact on the attack parameters, and will be removed in our formal experiments.

By testing all LESS parameters, we found that each loop iteration takes around 20 cycles on our device. Therefore, we set the attack parameter with $\mathsf{ext\_offset}$ in $[80, 180]$, width in $[6, 25]$ and offset in $[-12, 12]$ and recorded the number of successful attack offset values under different $(\mathsf{ext\_offset}, \mathsf{width})$ pairs, some test results are shown in Figure 2.

From Figure 2, we observe that successful attacks cluster in the range $6 \leq \mathsf{width} \leq 13$. Given that a single iteration consumes c clock cycles, and we identified a set of attack parameter triples $(\mathsf{ext\_offset}, \mathsf{width}, \mathsf{offset})$ that achieve a high success rate when targeting the i^*-th iteration, the modified triple $(\mathsf{ext\_offset} + c, \mathsf{width}, \mathsf{offset})$ will likewise yield a high attack success rate for the $(i^* + 1)$-th iteration. As the appropriate $\mathsf{ext\_offset}$ value can be determined here, we can fix it and enumerate the $(\mathsf{width}, \mathsf{offset})$ pair more precisely in the test to find the (approximate) optimal parameters to perform our attack.



(a) $(n, t) = (252, 68)$, $c = 23$ (b) $(n, t) = (252, 192)$, $c = 20$ (c) $(n, t) = (548, 345)$, $c = 19$

Figure 2: Success attack count with different $\mathsf{ext\_offset}$ and width

It is critical to note that in the "Single Iteration Skip" model, the ideal attack parameter should maximize the conditional probability Pr[Skip the iteration | $b_{i^*} \neq 0$] rather than the marginal probability Pr[Skip the iteration]. This is because skipping the iteration at index i^* when $b_{i^*} = 0$ is operationally meaningless. However, the "Loop Abort" model is immune to this effect, as all subsequent iterations will not be executed after the i^*-th iteration.

4.4 Identifying Effective Leakage Information

In practical fault injection attacks, the fault injection process is not always successful, which produces irrelevant data that interferes with the output of Algorithm 3. Therefore, identifying valid faulty signatures from the large quantities of collected samples to enable effective key recovery also an important challenge.

In Algorithm 3, for any two leakage information $(V_0, R_0), (V_1, R_1)$ that extracted from Algorithm 2, if they are all relevant to the permutation of the same secret matrix $\mathbf{Q}_{s^*}$, then for any set $\mathsf{Candidate}_i$, the following must hold (the universal set is $\mathbb{Z}_n$):

$$\mathsf{Candidate}_i \in \{V_0 \cap V_1,\ V_0^c \cap V_1,\ V_0 \cap V_1^c,\ V_0^c \cap V_1^c\} \tag{14}$$

From (14), for all sets in the $\mathsf{Candidate}$ array, and for any distinct integers $i, j \in \mathbb{Z}_n$, either $\mathsf{Candidate}_i = \mathsf{Candidate}_j$ or $\mathsf{Candidate}_i \cap \mathsf{Candidate}_j = \emptyset$. By leveraging this property, we proposed Algorithm 4 to validate the collected information and discard all invalid data with retaining the majority of related (V, R) pairs. This algorithm is based on the disjoint set union with time complexity $O(m^2 n)$. Here, m denotes the number of (V, R) tuples collected.

For the scheme parameters with $s > 2$, if we only take the largest set in the DSU, we may require an excessive amount of leakage information to recover all the secret monomial matrices, leading to the loss of numerous useful leakages. Therefore, a threshold value K (preferably between 3 and 5) can be introduced in Algorithm 4, which returns all sets in DSU whose size exceeds K and clusters them by the number of their occurrences. This modification can effectively reduce the leakage loss and enable a more efficient recovery process for the permutation of these secret matrices.

5 Attack Results

5.1 Attack Result in Single Iteration Skip Model

For the "Single Iteration Skip" model with zero-filled $\mathbf{f}$, we performed the attack at $i = 4$ on ChipWhisperer, and the results are presented in the first seven rows of Table 1. According to the results, for all the seven scheme parameters, the average number of signing oracle accesses required in our attacks is consistently below 300.

To evaluate the attack success rate across different target indices, we carried out experiments on distinct index positions within the target loop. Beyond the baseline target index $i = 4$, we additionally selected four supplementary target indices for each protocol parameter, which were positioned at $0.25t$, $0.5t$, $0.75t$, and near the termination of the `for` loop. The experimental results are summarized in Table 2: For some scheme parameters, changing the target index has no significant influence on the success rate. In contrast, we observed that some specific parameters yielded a higher success rate if we change a target index. Accordingly, we conducted follow-up attacks targeting these high-success-rate indices, which effectively reduced the numbers of signing oracle accesses. The corresponding data are detailed in the last three rows of Table 1.

For the "Single Iteration Skip" model, if array $\mathbf{f}$ is statically addressed, the theoretical attack can be performed as follows: We first access the signing oracle to obtain a normal

Algorithm 4 Extracting valid leakage from collected data in Algorithm 2

Input: Public parameters (n, k, t, w, s), m collections $\{(V_0, R_0), (V_1, R_1), ..., (V_{m-1}, R_{m-1})\}$ from Algorithms 2 related to $\mathbf{Q}_{s^*}$

Output: The filtered information (V, R) as the input of Algorithm 3.

$\mathsf{DSU} \leftarrow \mathsf{MakeDSU}(m)$
for $i = 0$ to $m - 1$ **do**
  if $\mathsf{DSU.Find}(i) \neq i$ **then**
    continue
  end if
  Initialize $\mathsf{Candidate}_j \leftarrow \mathbb{Z}_n$ for $j = 0$ to $n - 1$.
  for $j = i + 1$ to $m - 1$ **do**
    if $\mathsf{DSU.Find}(j) \neq j$ **then**
      continue
    end if
    for t in $\{i, j\}$ **do**
      for $u = 0$ to $n - 1$ **do**
        if $u \in R_t$ **then**
          $\mathsf{Candidate}_u \leftarrow \mathsf{Candidate}_u \cap V_t$
        else
          $\mathsf{Candidate}_u \leftarrow \mathsf{Candidate}_u \cap V_t^c$
        end if
      end for
    end for
    $\mathsf{CandidateS} \leftarrow \{S \mid S \in \mathsf{Candidate}\}$ //CandidateS has four elements.
    Check consistency: each $\mathsf{CandidateS}_u$ has exactly $|\mathsf{CandidateS}_u|$ copies in Candidate
    Check disjointness: all $\mathsf{CandidateS}_u \cap \mathsf{CandidateS}_v = \emptyset$ for $u, v \in \mathbb{Z}_4, u \neq v$
    if checks pass **then**
      $\mathsf{DSU.Union}(i, j)$
    end if
  end for
end for
Take the largest subset S_{max} from DSU.
return $\{(V_i, R_i) | \ i \in S_{\mathsf{max}}\}$ as the input of Algorithm 3

signature and record the indces where $b_i = 0$ (corresponds to $f_i = 0$). Subsequently, we immediately perform an adaptive fault injection attack targeting one of these recorded indices. If the attack success rate is maintained, the required number of signing oracle accesses will be twice that of the "zero-filled" case.

5.2 Attack and Simulation Result in Loop Abort Model

5.2.1 Attack Result in Zero-Filled Array Case

In the "Loop Abort" model, we noticed the root of GGM-Tree is forced not to be published in LESS, so we have to avoid letting $\mathbf{f}$ contain all zeros as no seeds will be published then. Therefore, we conducted the experiment by skipping out of the loop at $i = 20$ if the challenge is sparse, otherwise, we skipped out at $i = 4$. The experiment results are shown in Table 3.

From the attack results, it can be observed that the number of signing oracle accesses is significantly reduced in comparison with the "Single Iteration Skip" model. For some parameters, only a single access suffices to recover the secret key. Although we were unable to identify high-success-rate attack parameters for the scheme parameter $(n, t, w, s) = (252, 68, 42, 4)$ and $(400, 102, 61, 4)$, we still recovered the secret key with an average of fewer than 50 and 20 signing oracle accesses, respectively.

Table 1: Attack result in "Single Iteration Skip" model (zero-filled $\mathbf{f}$)

(n, t, w, s)	Skip index	Success rate	Trials	Min	Max	Average
$(252, 192, 36, 2)$	4	18.32%	32	44	165	83.59
$(252, 68, 42, 4)$	4	47.94%	32	95	261	156.81
$(252, 45, 34, 8)$	4	71.58%	32	167	307	223.94
$(400, 220, 68, 2)$	4	31.51%	32	32	105	60.53
$(400, 102, 61, 4)$	4	51.03%	32	94	170	131.88
$(548, 345, 75, 2)$	4	100.00%	32	48	132	85.13
$(548, 137, 79, 4)$	4	46.84%	32	102	210	144.09
$(252, 68, 42, 4)$	32	70.50%	32	79	199	121.34
$(400, 102, 61, 4)$	50	90.28%	32	54	103	74.16
$(548, 137, 79, 4)$	68	95.29%	32	62	181	82.34

Table 2: Success rate with skipping at different indices

(n, t, w, s)	Index 1	Rate 1	Index 2	Rate 2	Index 3	Rate 3	Index 4	Rate 4
$(252, 192, 36, 2)$	50	22.2%	96	23.4%	142	20.2%	188	20.6%
$(252, 68, 42, 4)$	18	45.6%	32	70.5% ↑	46	55.1%	60	55.1%
$(252, 45, 34, 8)$	13	79.7%	22	75.1%	31	73.4%	40	75.5%
$(400, 220, 68, 2)$	57	29.8%	110	29.2%	163	31.1%	216	30.3%
$(400, 102, 61, 4)$	27	48.4%	50	90.2% ↑	73	63.6%	96	58.4%
$(548, 345, 75, 2)$	88	100.0%	172	100.0%	256	20.0% ↓	340	100.0%
$(548, 137, 79, 4)$	36	45.3%	68	95.3% ↑	100	57.9%	132	56.8%

5.2.2 Simulation Result in Statically-Addressed Array Case

In this case, we performed simulations on Ubuntu 25.10 and set the success rate to be 100%. Notably, we did not account for information loss during the extraction, collection, and filtering processes. Leveraging the static property of array $\mathbf{f}$, we adopted the strategy of executing one attack after each normal execution. Though a single normal execution can be followed by multiple attacks to achieve higher efficiency, the inherent uncertainty of fault injection introduces unpredictability into the state of array $\mathbf{f}$. Corresponding simulation results are summarized in Table 4.

6 Theoretical Analysis

6.1 Leakage Demand Analysis

After checking the experiment result, we can theoretically analyze the number of leakages and signing oracle accesses demanded to recover all secret matrices and give an approximate result.

6.1.1 Leakage Demand for Single Secret Matrix Recovery

For a particular secret matrix $\mathbf{Q}_{i^*}$, if we obtain its permutation, then its coefficients can be directly recovered. Then, the number of leakages needed to recover the secret matrix is just the number of leakages required to recover its permutation, and we can obtain the following theorem:

Theorem 1. *Recovering the permuation of a single private monomial matrix $\mathbf{Q}_{i^*}$ demands approximate $2\log_2 n$ leakages related to it.*

Proof. This problem can be abstracted as the following "labelling" problem: Suppose there are $n = 2k \geq 252$ different elements, and we randomly label k elements with `0` and

Table 3: Attack result in "Loop Abort" model (zero-filled $\mathbf{f}$)

(n, t, w, s)	Index	Success Rate	Trials	Min	Max	Average	Normalized
$(252, 192, 36, 2)$	20	100.00%	400	1	3	1.03	1.03
$(252, 68, 42, 4)$	4	10.67%	130	4	134	46.35	4.95
$(252, 45, 34, 8)$	4	100.00%	181	5	24	10.76	10.76
$(400, 220, 68, 2)$	20	100.00%	266	1	1	1.00	1.00
$(400, 102, 61, 4)$	4	12.87%	312	1	61	14.58	1.88
$(548, 345, 75, 2)$	20	100.00%	209	1	2	1.01	1.01
$(548, 137, 79, 4)$	20	100.00%	231	1	3	1.55	1.55

Table 4: Simulation result in "Loop Abort" model (statically-addressed $\mathbf{f}$)

(n, t, w, s)	Trials	Min	Max	Average
$(252, 192, 36, 2)$	985	2	4	2.02
$(252, 68, 42, 4)$	280	6	14	9.23
$(252, 45, 34, 8)$	113	30	82	43.73
$(400, 220, 68, 2)$	990	2	4	2.00
$(400, 102, 61, 4)$	292	4	10	6.79
$(548, 345, 75, 2)$	344	2	4	2.01
$(548, 137, 79, 4)$	284	4	10	5.67

the remaining k elements with `1`. We perform the procedure T times, and each element's label sequence can be written as a vector $\mathbf{s}_i \in \{\mathtt{0}, \mathtt{1}\}^T$.

Since exactly k elements are labelled `0` and the remaining k elements are labelled `1` in each round of labelling, the probability that any two distinct elements i, j receive different labels in a single round is:

$$p_{\text{diff}} = \frac{2 \cdot \mathrm{C}_{n-2}^{k-1}}{\mathrm{C}_n^k} = \frac{2k^2}{n(n-1)} \overset{2k=n}{=\!=} \frac{n}{2(n-1)} \overset{n \geq 252}{\approx} \frac{1}{2}. \tag{15}$$

As each labelling process is independent, the probability that the $\mathbf{s}_i = \mathbf{s}_j$ retains after T times labelling iterations is $(1 - p_{\text{diff}})^T \approx 2^{-T}$. We require the expected number of collisions to be less than 1 to guarantee valid recovery with high probability. As there are $\mathrm{C}_n^2 = \frac{n^2-n}{2}$ different (i, j) pairs, the expected number of collisions is $\frac{n^2-n}{2} \cdot 2^{-T}$. Therefore, we obtain:

$$\frac{n^2 - n}{2} \cdot 2^{-T} < 1 \implies T > 2\log_2 n - 1 + \log_2\left(1 - \frac{1}{n}\right) \approx 2\log_2 n - 1. \tag{16}$$

Thus, the expected number of leakage instances required is about $T \approx 2\log_2 n$. For the convenience of subsequent analysis, we take its ceiling value $\lceil 2\log_2 n \rceil$.

□

6.1.2 Leakage Demand for All Secret Matrices Recovery

Since we have $s - 1$ distinct secret matrices, each effective leakage corresponds to one of them. However, the average number of required effective leakages exceeds the intuitive value $2(s-1)\log_2 n$. This is because we may waste some accesses when some secret matrices can be fully recovered while others have not yet reached there required thresholds. Determining the actual expected value reduces to the multiple coupon collection problem [New60, FS14]: given N coupon types, we draw one random coupon per trial, aiming to collect at least M coupons of each type. We use $\mathsf{Cop}(N, M)$ to denote the expected number of trials needed for convenience. For $M \geq 2$, the exact value can be calculated with (17) [New60].

Table 5: $\mathsf{Cop}(s-1, \lceil 2\log_2 n \rceil)$ on LESS parameters

(n, t, w, s)	$s-1$	$\lceil 2\log_2 n \rceil$	$\mathsf{Cop}(s-1, \lceil 2\log_2 n \rceil)$
$(252, 192, 36, 2)$	1	16	16.00
$(252, 68, 42, 4)$	3	16	58.35
$(252, 45, 34, 8)$	7	16	152.44
$(400, 220, 68, 2)$	1	18	18.00
$(400, 102, 61, 4)$	3	18	64.97
$(548, 345, 75, 2)$	1	19	19.00
$(548, 137, 79, 5)$	3	19	68.27

Table 6: Theoretical trials in "Single Iteration Skip" model with zero-filled array $\mathbf{f}$

(n, t, w, s)	$\epsilon = 0.6$	$\epsilon = 0.8$	$\epsilon = 0.99$	Experiment
$(252, 192, 36, 2)$	142.22	106.67	86.20	83.59
$(252, 68, 42, 4)$	157.45	118.09	95.43	156.81
$(252, 45, 34, 8)$	336.26	252.20	203.80	223.94
$(400, 220, 68, 2)$	97.06	72.79	58.82	60.53
$(400, 102, 61, 4)$	181.06	135.80	109.74	131.88
$(548, 345, 75, 2)$	145.67	109.25	88.28	85.13
$(548, 137, 79, 4)$	197.32	147.99	119.59	144.09

$$\mathsf{Cop}(N, M) = \int_0^\infty \left(1 - \left[1 - \mathrm{e}^{-t} \sum_{k=0}^{M-1} \frac{t^k}{k!}\right]^N\right) \mathrm{d}t \tag{17}$$

For LESS parameters, we can use (17) to calculable all the $\mathsf{Cop}(s-1, \lceil 2\log_2 n \rceil)$ values, as demonstrated in Table 5.

6.2 Required Signing Oracle Accesses in Fault Models

In the previous section, we only considered the amount of leakage required to recover all secret matrices. However, after determining the required number of leakages, it is also necessary to evaluate the number of signing oracle accesses in different fault models.

6.2.1 Required Accesses in Single Iteration Skip Model

For the "Single Iteration Skip" model with zero-filled array $\mathbf{f}$, we selected attack parameters that maximize $\Pr[\text{successful skip} \mid b_i \neq 0]$. If we set this conditional probability to be ϵ, then the overall success rate is:

$$\Pr[\text{successful skip}] = \epsilon \cdot \Pr[b_i \neq 0] = \epsilon \cdot \frac{w}{t} \tag{18}$$

As array $\mathbf{f}$ contains all zeros in this case, each successful skip indicates one effective leakage. Therefore, the expression of expected number of accesses is:

$$\mathbb{E}_{11}[\text{accesses}] = \frac{\mathsf{Cop}(s-1, \lceil 2\log_2 n \rceil)}{\Pr[\text{successful skip}]} = \frac{t \cdot \mathsf{Cop}(s-1, \lceil 2\log_2 n \rceil)}{w \cdot \epsilon} \tag{19}$$

According to (19), we can calculate the expectation number of signing oracle accesses with different success-skip rate. The value is shown in Table 6, and it can be observed that the experiment result fits the formula well.

If array $\mathbf{f}$ is statically-addressed with no initial value guaranteed, the attacker need to obtain a normal signature before conducting an the attack. Therefore, compared to

the "zero-filled" case, twice accesses will be needed, and we obtain the expected access numbers to be:

$$\mathbb{E}_{12}[\text{accesses}] = \frac{2t \cdot \mathsf{Cop}(s-1, \lceil 2\log_2 n \rceil)}{w \cdot \epsilon} \tag{20}$$

6.2.2 Required Accesses in Loop Abort Model

With zero-filled array f: In the "Loop Abort" model, as one successful attack may trigger multiple leakages, we can calculate the expected number of leakages for one trial after the i^*-th iteration to be $(t-1-i^*)\cdot\frac{w}{t}$. However, if array $\mathbf{f}$ contains all zeros after the fault injection, no seed will be published. To derive the expected access numbers, we can calculate the probability of obtaining exact u leakages if we perform the fault injection attack after the i^*-th iteration $\mathsf{prob}_u(i^*)$:

$$\mathsf{prob}_u(i^*) = \frac{1}{\mathrm{C}_t^{t-i^*-1}} \cdot \begin{cases} \mathrm{C}_w^u \cdot \mathrm{C}_{t-w}^{t-i^*-1-u}, 1 \le i^* \le w-1 \\ \mathrm{C}_{t-w}^{t-i^*-1} + \mathrm{C}_{t-w}^{t-w-i^*-1}, u = 0 \text{ and } t-i^*-1 \ge w \\ \mathrm{C}_{t-w}^{t-i^*-1}, u = 0 \text{ and } t-i^*-1 < w \end{cases} \tag{21}$$

Therefore, for a fixed i^* we can calculate the expected number of accesses by the following recurrence relation. For simplicity, we consider the demanded number of leakages to be an integer $\lceil \mathsf{Cop}(s-1, \lceil 2\log_2 n \rceil) \rceil$.

$$\mathsf{rec}_\ell(i^*) = \begin{cases} \dfrac{1 + \sum_{j=1}^{w-1} \left(\mathsf{prob}_j(i^*) \cdot \mathsf{rec}_{(\ell+j)}(i^*)\right)}{1 - \mathsf{prob}_0(i^*)}, & \ell < \lceil \mathsf{Cop}(s-1, \lceil 2\log_2 n \rceil) \rceil \\ 0, & \text{otherwise} \end{cases} \tag{22}$$

Following (22), the value $\mathsf{rec}_0(i^*)$ is the expected effective accesses. Considering the success rate ϵ, we obtain (23) and its plot for the first three LESS parameters in Figure 3.

$$\mathbb{E}_{21}[\text{accesses}](i^*) = \frac{\mathsf{rec}_0(i^*)}{\epsilon} \tag{23}$$

With statically-addressed array f: In this case, array $\mathbf{f}$ is statically addressed with its initial values unknown, and the "one attack after a normal signature" strategy should be adapted. As array $\mathbf{f}$ definitely contains 1 after a normal signing process execution, there is no need to consider the situation where all-zero $\mathbf{f}$ leads to no seed publication. Here, the optimal attack point occurs when $i^* = 0$. This implies that only f_0 is updated by the process if the attack succeeds. The effective leakages in a successful attack is:

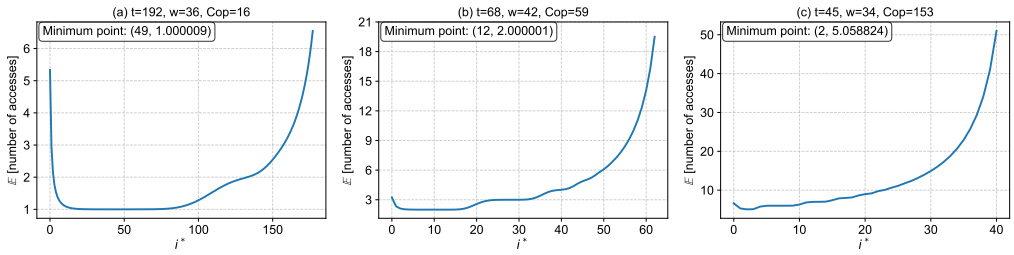


Figure 3: Plot of $\mathbb{E}_{21}[\text{accesses}](i^*)$ for LESS parameters with $n = 252, \epsilon = 1$

Table 7: Theoretical result in "Loop Abort" model with statically-addressed array

(n, t, w, s)	$\epsilon = 0.99$	Simulation	(n, t, w, s)	$\epsilon = 0.99$	Simulation
$(252, 192, 36, 2)$	2.02	2.02	$(400, 102, 61, 4)$	6.06	6.79
$(252, 68, 42, 4)$	8.08	9.23	$(548, 345, 75, 2)$	2.02	2.01
$(252, 45, 34, 8)$	38.38	43.73	$(548, 137, 79, 4)$	6.06	5.67
$(400, 220, 68, 2)$	2.02	2.00			

$$L = w \cdot \frac{t-1}{t} \cdot \frac{t-w}{t} = \frac{w \cdot (t-1) \cdot (t-w)}{t^2} \tag{24}$$

As performing an attack requires accessing the signing oracle twice, and the expected leakage demand is $\mathsf{Cop}(s-1, \lceil 2\log_2 n \rceil)$. Assuming a success rate of ϵ for a single attack, we obtain (25) as the the expected number of accesses. For the seven scheme parameters, corresponding $\mathbb{E}_{22}[\text{accesses}]$ values are displayed in Table 7.

$$\mathbb{E}_{22}[\text{accesses}] = \frac{2}{\epsilon} \left\lceil \frac{\mathsf{Cop}(s-1, \lceil 2\log_2 n \rceil)}{L} \right\rceil = \frac{2}{\epsilon} \left\lceil \frac{t^2 \cdot \mathsf{Cop}(s-1, \lceil 2\log_2 n \rceil)}{w \cdot (t-1) \cdot (t-w)} \right\rceil \tag{25}$$

7 Conclusion

In this work, we presented a novel fault injection attack targeting LESS. Unlike existing fault attacks that focus on the complex GGM-Tree structure, our approach exploits vulnerabilities inherent in the computation process of the binary array $\mathbf{f}$. Through extensive cross-device testing, we have identified two common initial state cases for array $\mathbf{f}$ and proposed two fault models. Our attack achieves bidirectional derivation of attack parameters and fault position, which both eliminating for heuristic guessing of the fault position and enhancing the attacker's adaptability. Based on this, we designed a series of supporting algorithms and performed the attack experiments and simulations to confirm the high efficiency of our approach. Beyond experimental validation, we also provide a theoretical analysis of the expected number of signing oracle accesses required for successful recovery.

To prevent similar attacks, we suggest that for temporary arrays in the scheme be defined as pointer variables and use `malloc` function for memory allocation in the C implementation to enhance the randomness of temporary array addresses and contents within each function call. Additionally, some easier defense strategies like introducing a verification process with a time complexity of $O(t)$ before returning the signature result, or simply use $\mathbf{b}$ instead of $\mathbf{f}$ to calculate which seeds to be published. In the future research, we intend to generalize our attack framework to additional post-quantum cryptosystems, with a systematic evaluation of how such `for`-loop perturbation attacks may affect their physical security in real applications.

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